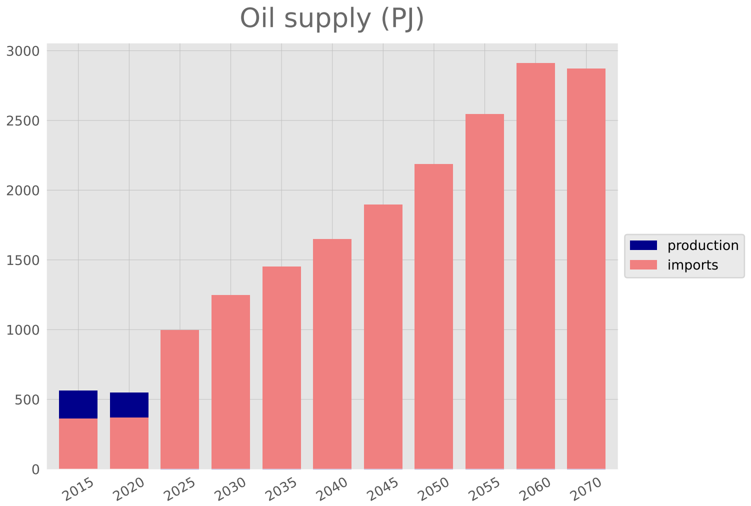
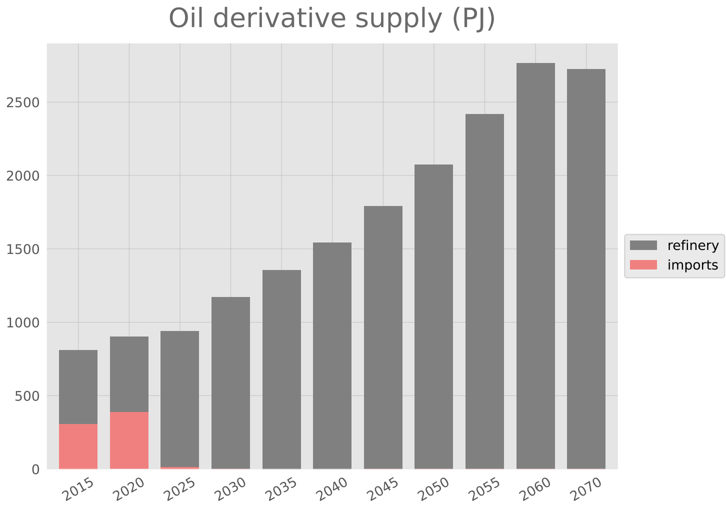
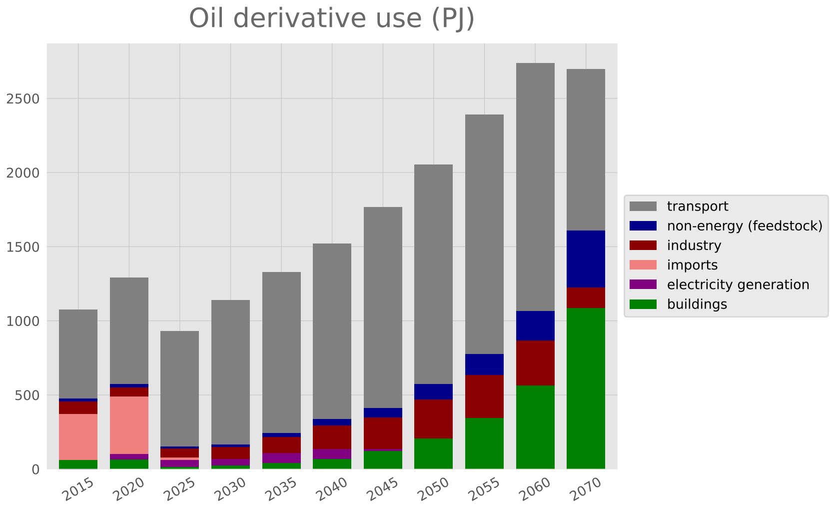
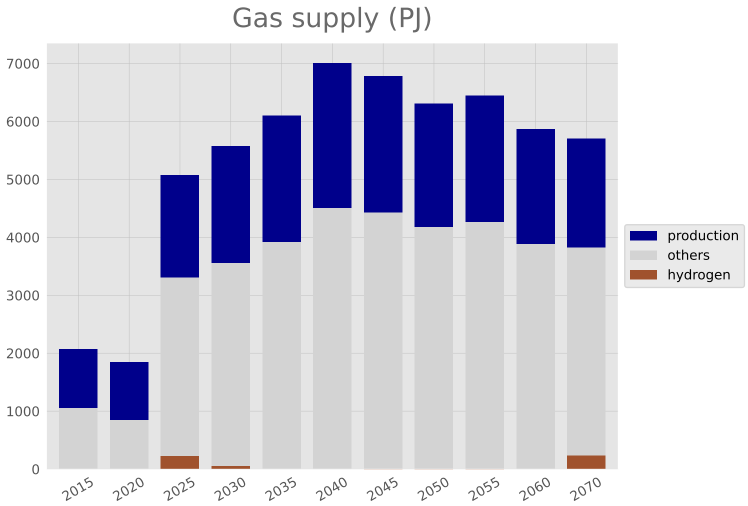


Oil supply (PJ)
3000
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
production
imports

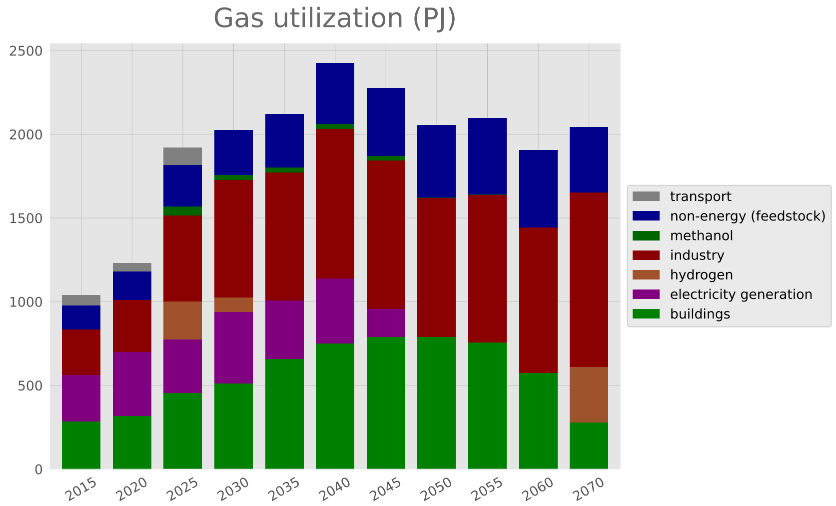

Oil derivative supply (PJ)
0
500
1000
1500
2000
2500
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
refinery
imports



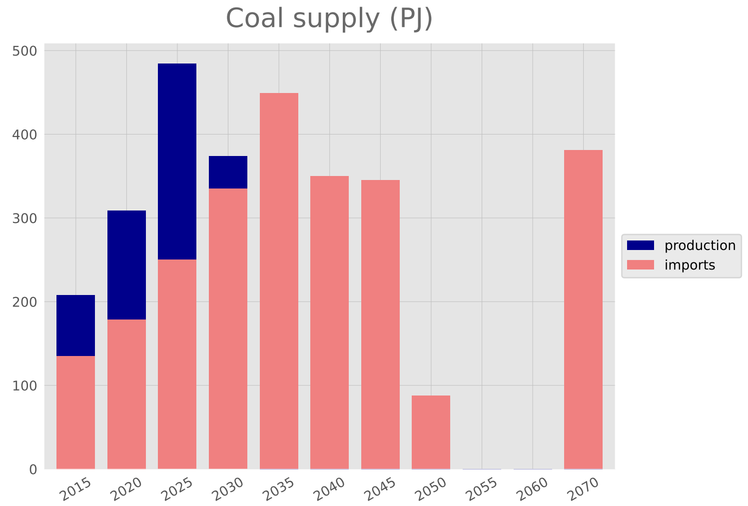
Oil derivative use (PJ)
0
500
1000
1500
2000
2500
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
non-energy (feedstock)
industry
imports
electricity generation
buildings

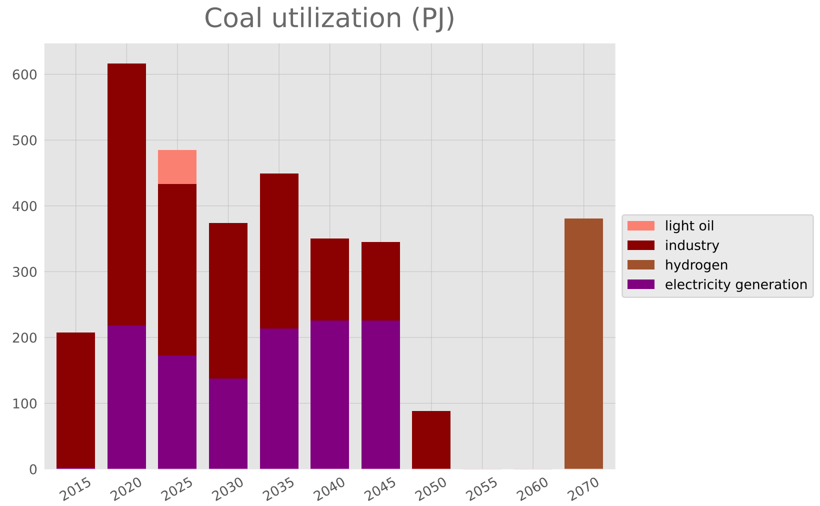


Gas supply (PJ)
7000
6000
5000
4000
3000
2000
1000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
production
others
hydrogen

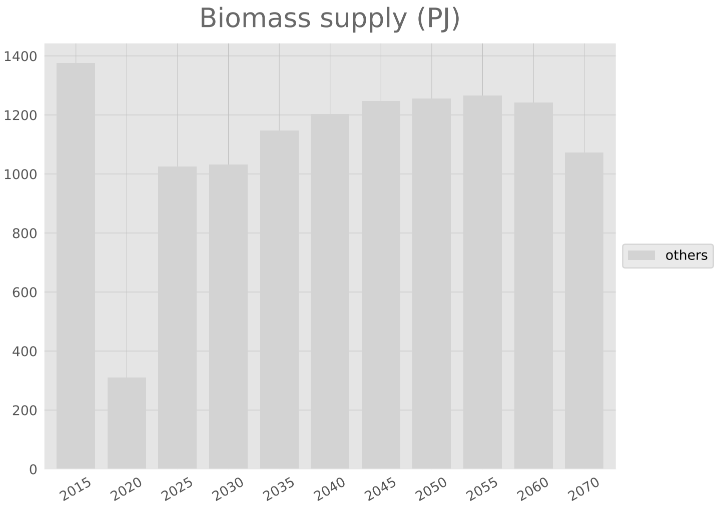


Gas utilization (PJ)
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
non-energy (feedstock)
methanol
industry
hydrogen
electricity generation
buildings

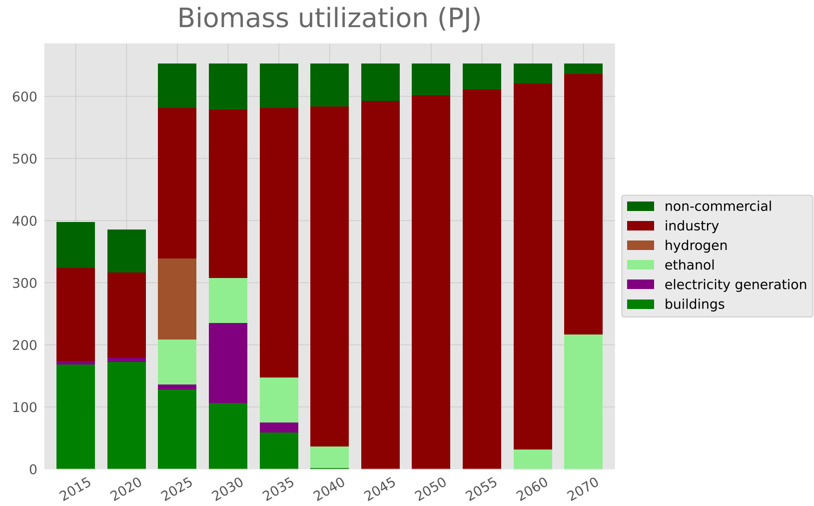

Coal supply (PJ)
500
400
300
200
100
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
production
imports



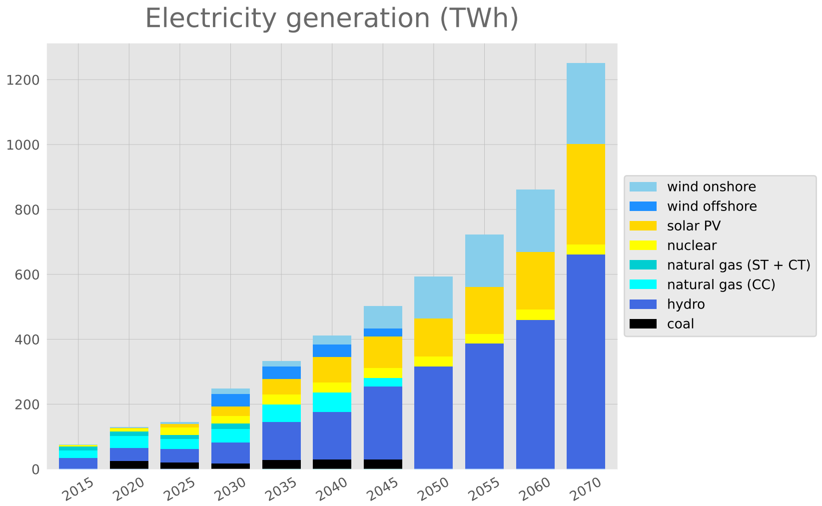
Coal utilization (PJ)
0
100
200
300
400
500
600
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
industry
hydrogen
electricity generation

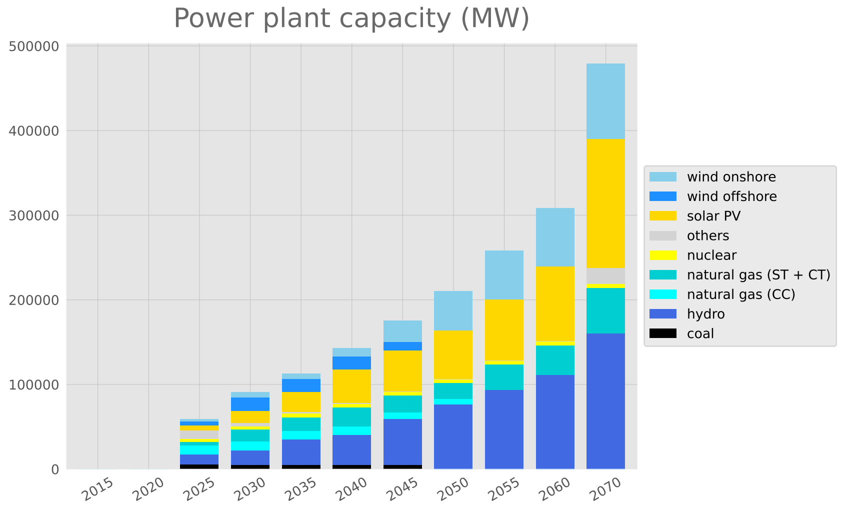


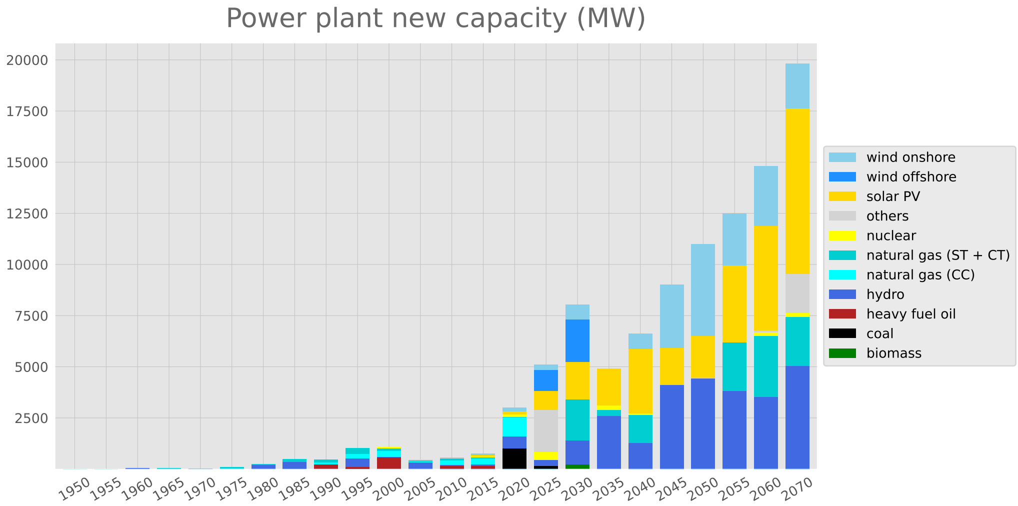
Biomass supply (PJ)
1400
1200
1000
800
600
400
200
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
others



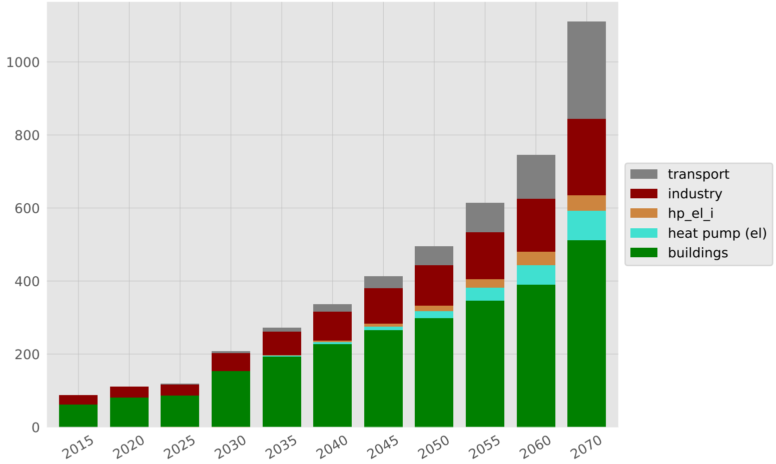
Biomass utilization (PJ)
0
100
200
300
400
500
600
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
non-commercial
industry
hydrogen
ethanol
electricity generation
buildings

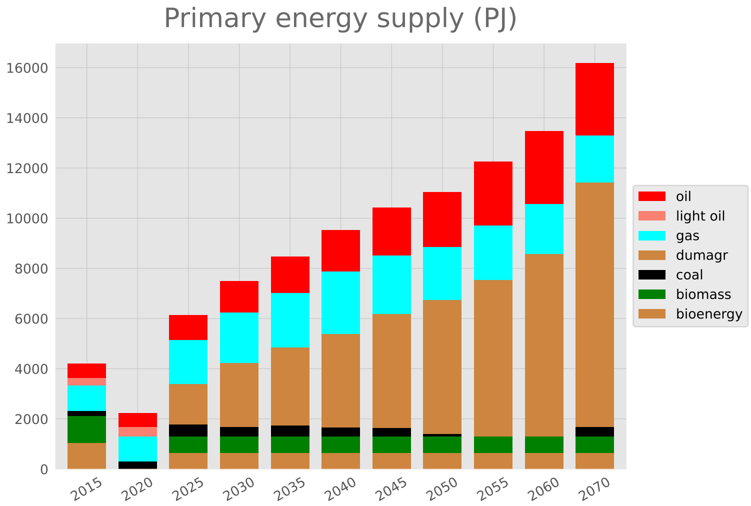

Electricity generation (TWh)
0
200
400
600
800
1000
1200
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
wind onshore
wind offshore
solar PV
nuclear
natural gas (ST + CT)
natural gas (CC)
hydro
coal

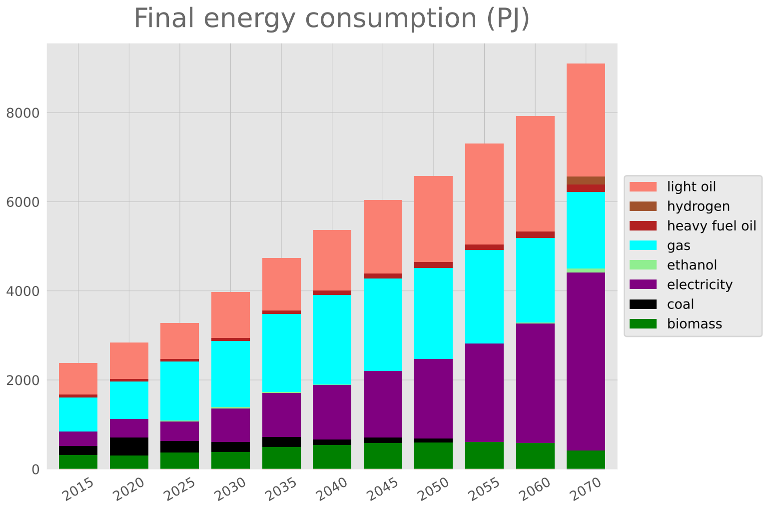

Power plant capacity (MW)
500000
400000
300000
200000
100000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
wind onshore
wind offshore
solar PV
others
nuclear
natural gas (ST + CT)
natural gas (CC)
hydro
coal


Power plant new capacity (MW)
20000
17500
15000
12500
10000
7500
5000
2500
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
wind onshore
wind offshore
solar PV
others
nuclear
natural gas (ST + CT)
natural gas (CC)
hydro
heavy fuel oil
coal
biomass

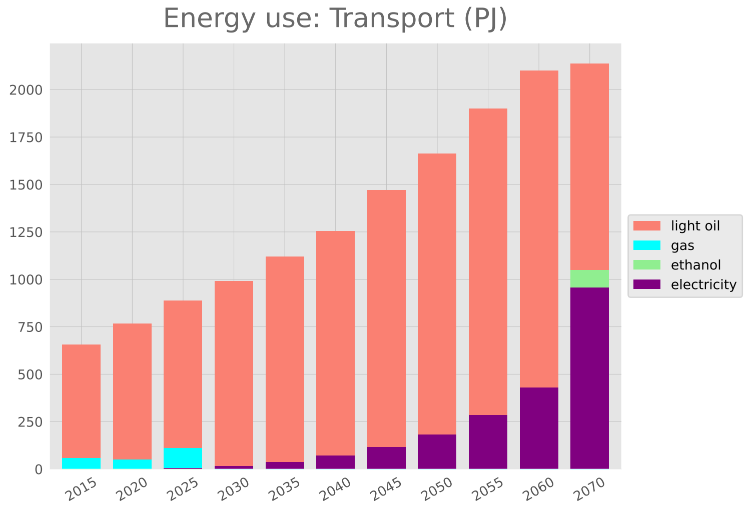
Electricity use (TWh)

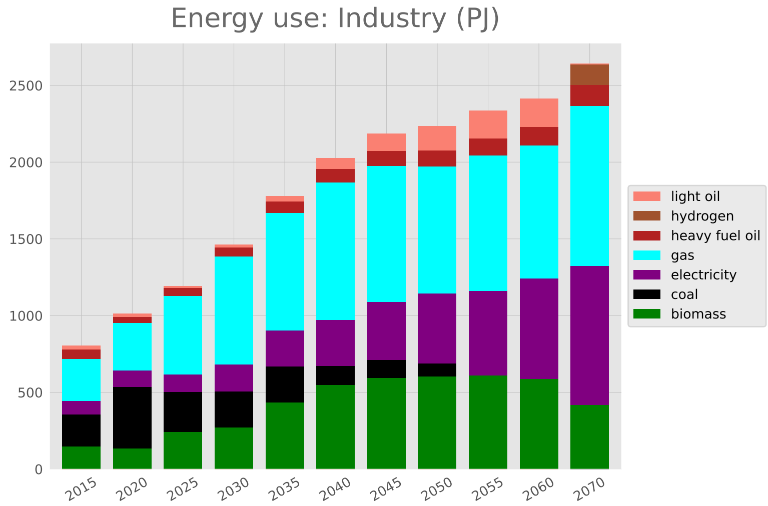


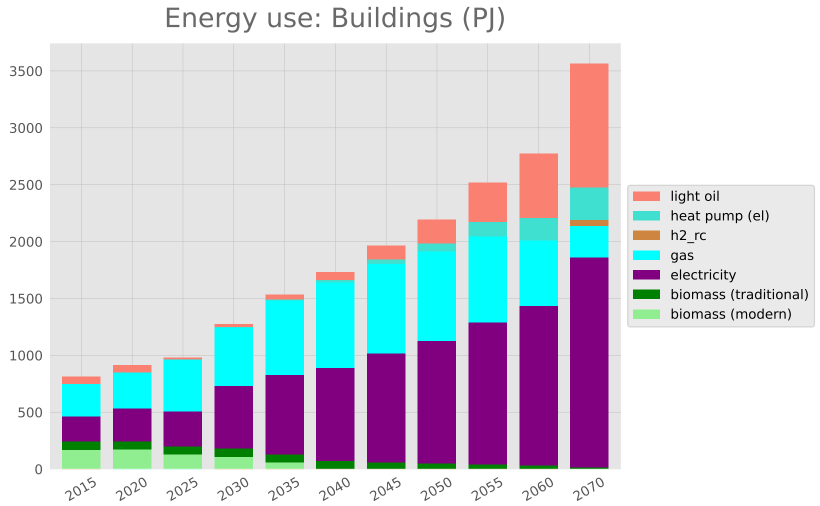

Primary energy supply (PJ)
16000
14000
12000
10000
8000
6000
4000
2000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
oil
light oil
gas
dumagr
coal
biomass
bioenergy



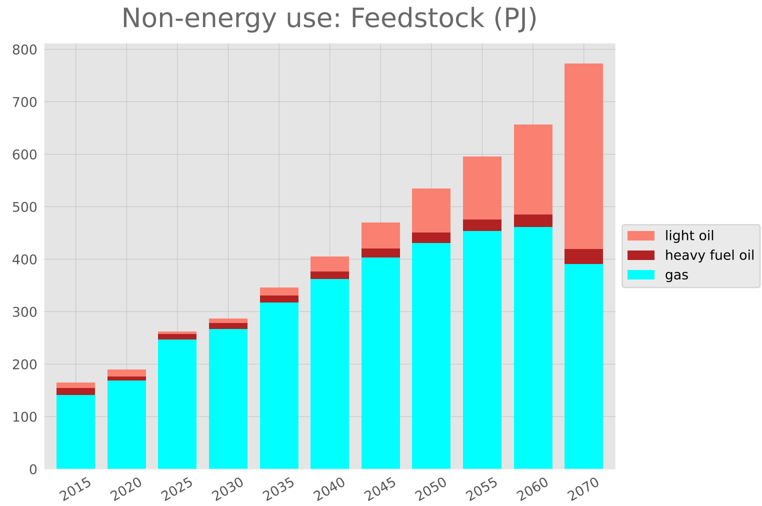
Final energy consumption (PJ)
0
2000
4000
6000
8000
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
hydrogen
heavy fuel oil
gas
ethanol
electricity
coal
biomass

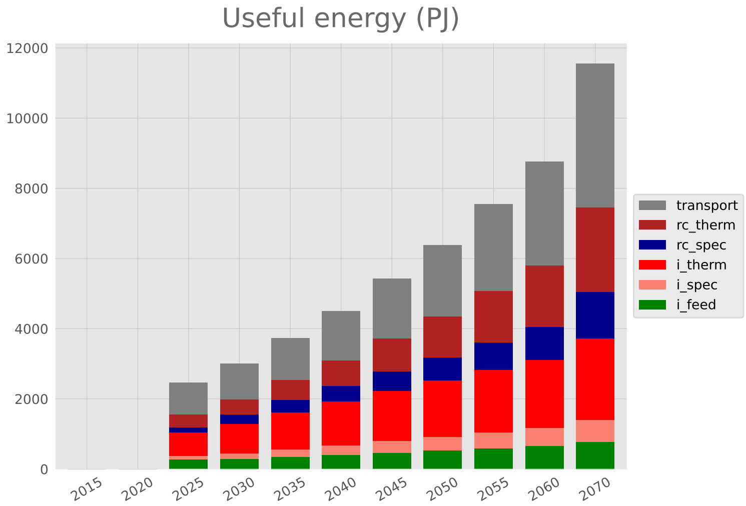

Energy use: Transport (PJ)
0
250
500
750
1000
1250
1500
1750
2000
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
gas
ethanol
electricity

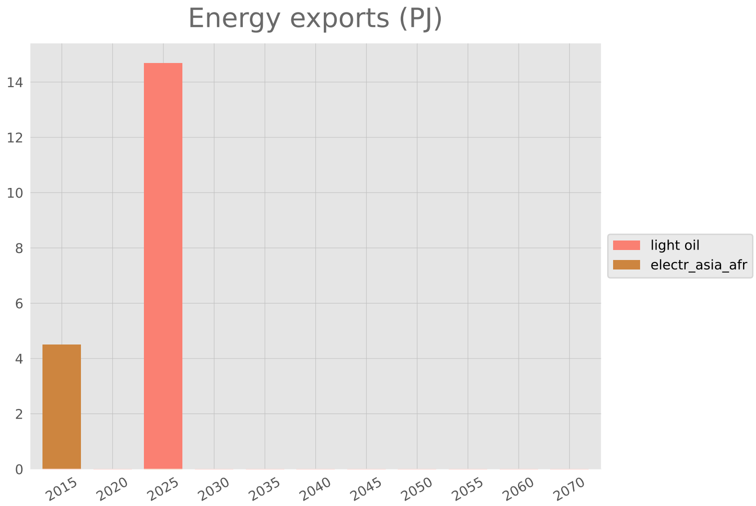

Energy use: Industry (PJ)
0
500
1000
1500
2000
2500
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
hydrogen
heavy fuel oil
gas
electricity
coal
biomass



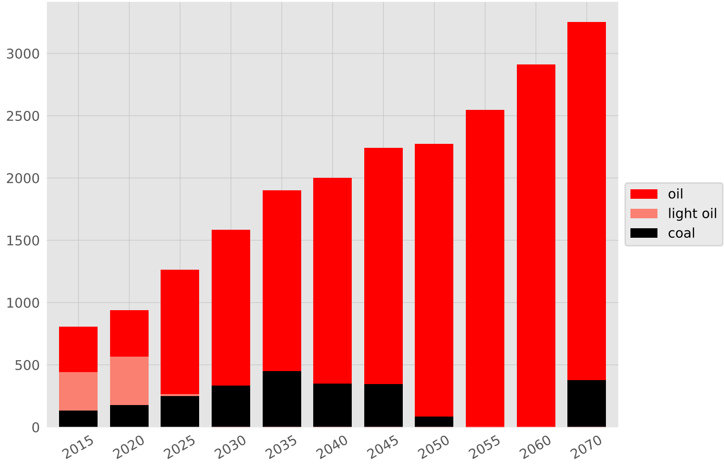
Energy use: Buildings (PJ)
3500
3000
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
heat pump (el)
h2_rc
gas
electricity
biomass (traditional)
biomass (modern)

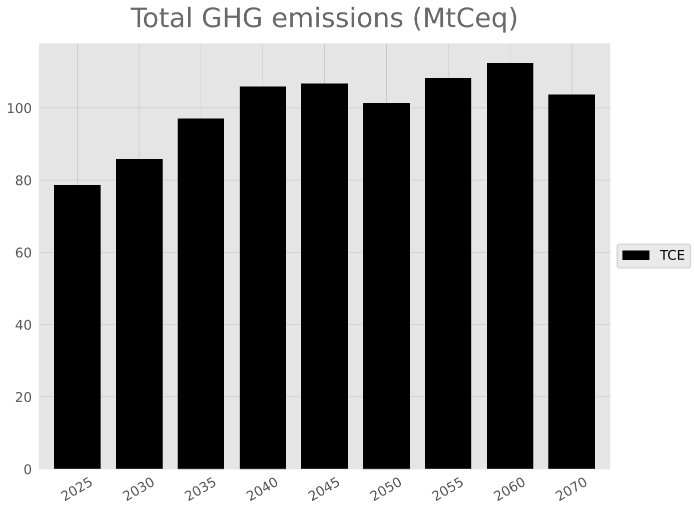

Non-energy use: Feedstock (PJ)
800
700
600
500
400
300
200
100
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
heavy fuel oil
gas


Useful energy (PJ)
12000
10000
8000
6000
4000
2000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
rc_therm
rc_spec
i_therm
i_spec
i_feed


Energy exports (PJ)
0
2
4
6
8
10
12
14
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
electr_asia_afr

Energy imports (PJ)




Total GHG emissions (MtCeq)
100
80
60
40
20
0
2025
2030
2035
2040
2045
2050
2055
2060
2070
TCE